



Getting Started with GCP

In this tutorial, you will learn how to set up a Google Cloud account, create a project, add your GCP educational credits to your account and project, and share your project with the instructor.

Log in to Google Cloud Console

Open an internet browser. If you have already logged in with a Gmail account, log out or use an incognito window. Type <https://console.cloud.google.com/> to access GCP Console. You need to login with your William & Mary Google Workspace Account (WMUsername@email.wm.edu). After then, you will be prompted with W&M's standard Microsoft login. You will login with your William & Mary email address (WMUsername@wm.edu) and password and complete the Duo Verification process

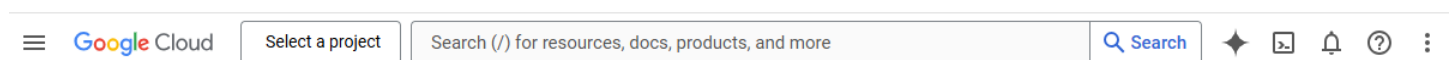
The first time you log in, you will get a welcome message that asks you to check the box that states that you agree to the terms and services.

After you agree and continue, you will see a screen like below that has a list of GCP products as well as popular resources to get started.

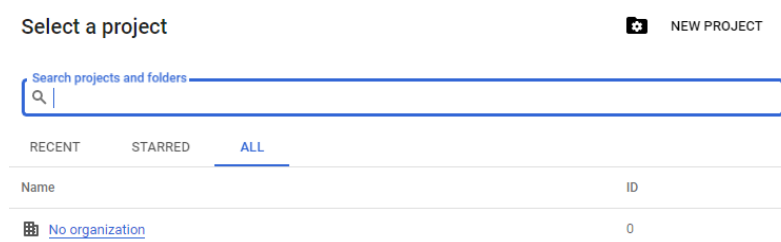
Create a Project

A project is a container for resources that you create and use on GCP. When you create a project, you are essentially creating a unique environment where you can deploy, manage, and use various GCP services.

In order to create a new project, click on *Select a project* at the top-left of the page.



To create a new project, click on *New Project*.





In the dialog box, you need to enter a project name. A project name can be anything, and it will prompt you if the project name you entered does not fulfill the name criteria. Please note that I used my WM Username (adurukansonmez) when creating the project. Your project name should be: WMusername-5132-project.

Every time you create a project, you name your project as you wish and at the same time, a universally unique Project ID is created and assigned to your project. If your project name was never used before, it will be assigned as the Project ID. Otherwise, GCP will create and assign a unique Project ID to your project. You have the option to edit it if you like.

You also have the option to link your project to an organization. If you used your William and Mary email account to log in, you should see wm.edu or email.wm.edu. If you used your Gmail account, you would not have that option, so you would leave it as "No Organization".

New Project

Project name *

adurukansonmez-5132-project

?

Project ID: adurukansonmez-5132-project. It cannot be changed later. [EDIT](#)

Organization *

email.wm.edu

▼

?

Select an organization to attach it to a project. This selection can't be changed later.

Location *

 email.wm.edu

[BROWSE](#)

Parent organization or folder

CREATE

CANCEL

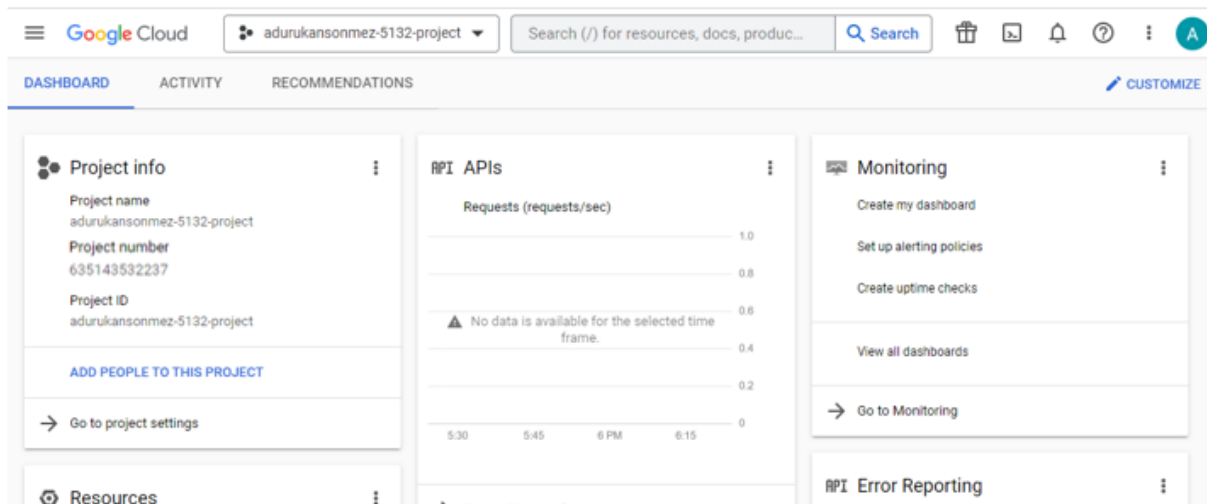
After you hit *Create*, it takes some time to create the project.

Notifications

 Create Project: adurukansonmez-5132-project Just now

[SELECT PROJECT](#)

You will either get a notification to select the project, or it will display on the screen directly. Either way, you will see your dashboard for this project, as shown below:



This dashboard is fairly empty; there is no errors, nothing to track or monitor since we just created it. As we use GCP tools and applications, it will include some monitoring and some tools that will help you see all of your resources at a glance.

On the top left portion of the dashboard, we have the project info, which includes project name, project number and project ID. When you use a GCP tool, you need to provide the Project ID, and this is one of the places you can find it.

Create a Billing Account and Link It to Your Project

As you know, using GCP costs money and companies have to pay GCP to use their tools, platforms, and services. However, in this course, we will use free credits provided by GCP for educational purposes.

In order to retrieve the coupon code for the education credits, you need to click on the link [Student Coupon Retrieval Link](#) and you will see the following form to fill out.

Enter your First and Last Name and use your WM email address. You need to use @wm.edu, not email.wm.edu as that domain is not programmed to receive emails.



Cloud Platform Education Grants

Use credits provided to you via the Google Cloud Platform Education Grants program to access Google Cloud Platform. Get what you need to build and run your apps, websites and services.

Thank you for your interest in Google Cloud Platform Education Grants. Please fill out the form below to receive a coupon code for credit to use on Google Cloud Platform.

First Name

Last Name

School Email

@wm.edu ▼

If you do not see your domain listed, please contact your course instructor: adurukansonmez@wm.edu

By clicking "Submit" below, you agree that we may share the following information with your educational institution and course instructor (adurukansonmez@wm.edu): (1) personal information that you provide to us on this form and (2) information regarding your use of the coupon and Google Cloud Platform products.

Submit

Once you fill out and submit the education grant, you will receive an email for verification. After verifying your email address, you will receive another email with your Coupon Code and Instructions. Click on the link to redeem your coupon code. Before you enter your coupon code, please make sure the email address is the one that you created your GCP account with.

GCP credit application

Fill in the following information below to apply GCP credits to your account listed below.

First name *

Ayse

Last name *

Durukan Sonmez

Account email

adurukansonmez@email.wm.edu

Credits will be applied to this account. If you'd like to apply credits to a different account, specify your preference [here](#).

Coupon code *

Terms and conditions

The following [Terms of Service](#) apply to the credit you received for Google Cloud products.

ACCEPT AND CONTINUE



After you enter your coupon code and continue, the educational credits will be applied to your account, and you will see the Billing Account Overview

Google Cloud

Search (/) for resources, docs, products, and more

Billing

Billing account: Billing Account for Education

Overview

Cost management

- Reports
- Cost table
- Cost breakdown
- Budgets & alerts
- Billing export

Cost optimization

- Committed use discounts (C...
- CUD analysis
- Pricing
- Cost estimation

Billing management

- Account management

Overview

BILLING ACCOUNT OVERVIEW

Current month
April 1 – 13, 2023

Month-to-date total cost
\$0.00

End-of-month total cost (forecasted)
Not enough historical data to project cost

View report

Keep track of your spend

Avoid surprises on your Google Cloud bill by creating budgets for your projects. Your billing administrators and users can be notified about how your spend tracks against your budget.

Why set up a budget alert

SET UP A BUDGET ALERT

Billing account: Manage
Billing Account for Education, 01C2FB-C08197-9369CD

Organization
email.wm.edu

Billing health checks

Check out your account health results to avoid common billing-related issues and adopt our best practice recommendations

Click on Account Management at the bottom of the Billing menu.

Google Cloud

Search (/) for resources, docs, products, and more

Billing

Billing account: Billing Account for Education

Overview

Cost management

- Reports
- Cost table
- Cost breakdown

Account management

CHANGE ORGANIZATION

RENAME BILLING ACCOUNT

CLOSE BILLING ACCOUNT

Billing account ID
01C2FB-C08197-9369CD

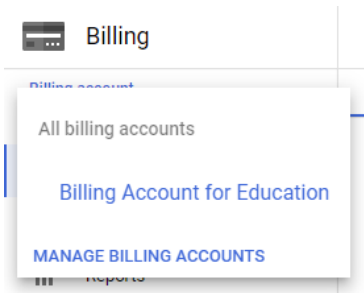
Organization
email.wm.edu

Projects linked to this billing account

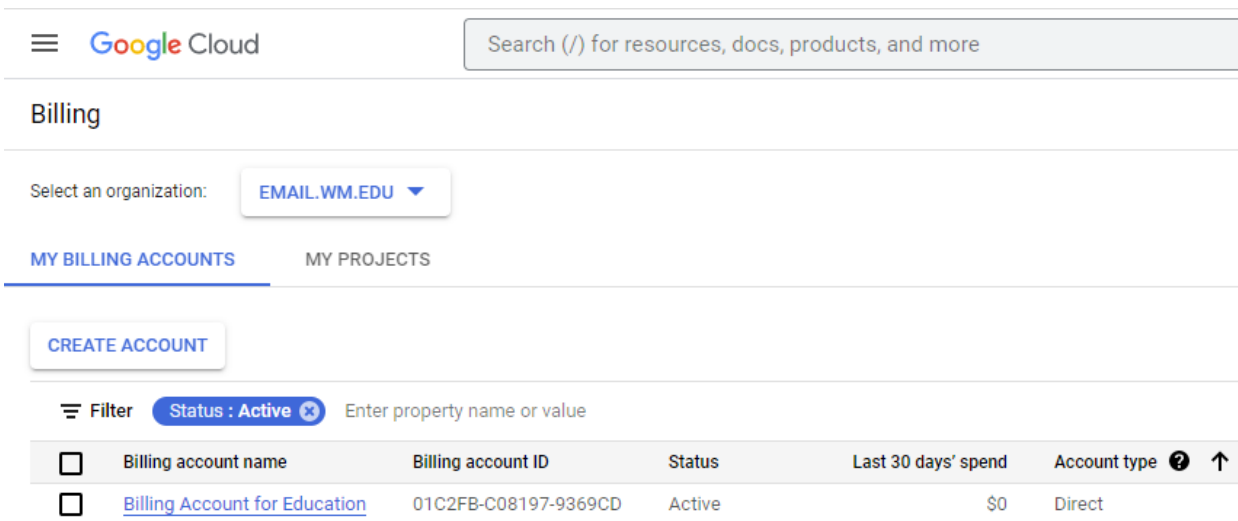
Project name	Project ID	Actions
There are no projects linked to this billing account.		

As you can see, there are no projects linked to the billing account.

When you click on the Billing Account for Education drop down menu, you will select the option "Manage Billing Accounts."



There, you will see a list of Billing Accounts and another tab for the projects.



Google Cloud

Search (/) for resources, docs, products, and more

Billing

Select an organization: EMAIL.WM.EDU

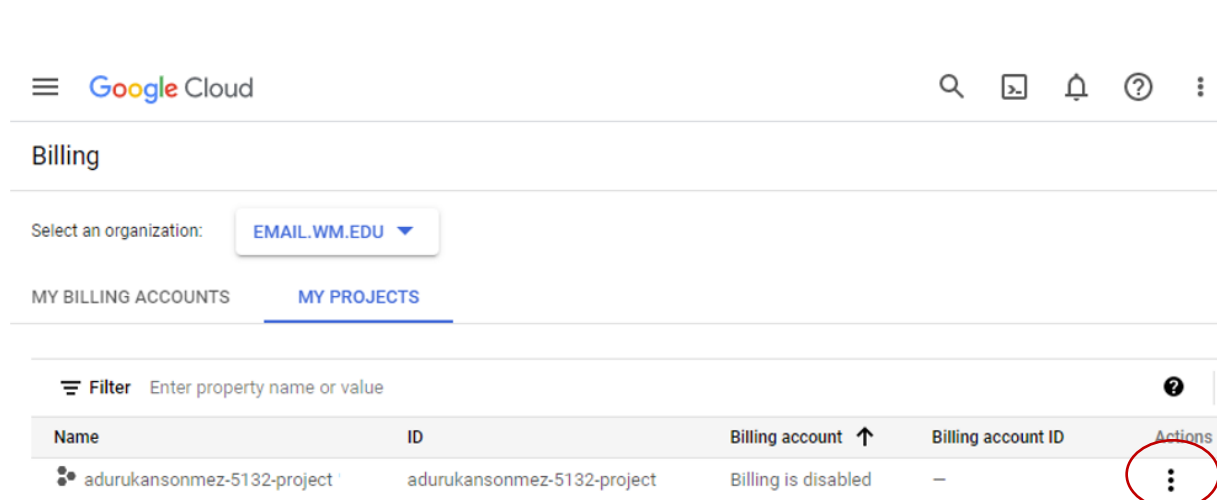
MY BILLING ACCOUNTS MY PROJECTS

CREATE ACCOUNT

Filter Status : Active Enter property name or value

Billing account name	Billing account ID	Status	Last 30 days' spend	Account type
Billing Account for Education	01C2FB-C08197-9369CD	Active	\$0	Direct

Click on the tab *My Projects*.



Google Cloud

Select an organization: EMAIL.WM.EDU

MY BILLING ACCOUNTS MY PROJECTS

Filter Enter property name or value

Name	ID	Billing account	Billing account ID	Actions
adurukansonmez-5132-project	adurukansonmez-5132-project	Billing is disabled	—	

Click on the actions menu icon, select Change Billing and in the drop-down menu below, select Billing Account for Education.



Set the billing account for project
"adurukansonmez-5132-project "

Billing account *
Billing Account for Education

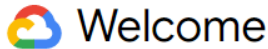
Any charges for this project will be billed to the account you select here.

CANCEL

SET ACCOUNT

Go back to Console by clicking on Google Cloud and then click on the dashboard.

Google Cloud email.wm.edu Search (/) for resources, docs, products, and more



You're working in email.wm.edu

[Dashboard](#) [Recommendations](#)

Create a VM

Run a query in BigQuery

Create a GKE cluster

Create a storage bucket

If the project you just created is listed on the Dashboard, you can select it. Otherwise, click on *Select Project* and then select it.

Google Cloud adurukansonmez-5132-project billing Search

DASHBOARD ACTIVITY RECOMMENDATIONS

Project info

Project name
adurukansonmez-5132-project

Project number
635143532237

Project ID
adurukansonmez-5132-project

ADD PEOPLE TO THIS PROJECT

Go to project settings

API APIs

Requests (requests/sec)

No data is available for the selected time frame.

Google Cloud Platform status

Google Compute Engine
Global: High Memory Issues on GCE Windows VMs using Google Cloud Ops Agent >= 2.27.0
Began at 2023-04-11 (15:28:29)
All times are US/Pacific
Data provided by status.cloud.google.com

Go to Cloud status dashboard

Monitoring



Share the Project with the Instructor

In order to make sure you have your project and other setup made correctly; I ask you to share your project with me. Unfortunately, Google Workspace for William & Mary has some issues with accessing the shared projects. Therefore, you will share your projects with an account I created for this course. Please note that this account is created for GCP purposes and emails sent to this account are not monitored.

On the Project Dashboard, click on *Add People to This Project* and add BUAD5132@gmail.com as an Owner.

Grant access to "adurukansonmez-5132-project "

Grant principals access to this resource and add roles to specify what actions the principals can take. Optionally, add conditions to grant access to principals only when a specific criteria is met. [Learn more about IAM conditions](#)

Resource

adurukansonmez-5132-project

Add principals

Principals are users, groups, domains, or service accounts. [Learn more about principals in IAM](#)

New principals

BUAD5132@gmail.com

Assign roles

Roles are composed of sets of permissions and determine what the principal can do with this resource. [Learn more](#)

Role *

Owner

IAM condition (optional)

+ ADD IAM CONDITION

Full access to most Google Cloud resources. See the list of included permissions.

+ ADD ANOTHER ROLE

SAVE

CANCEL